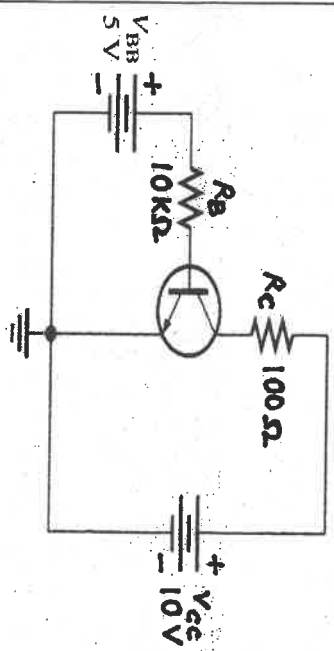


Total no. of Pages: Roll no.

II SEMESTER B.Tech.
MID TERM EXAMINATION, FEB.-2026
Paper Code – ECVT/IC ECC202 Subject: Electronic Devices & Circuits
Time: 1.5 Hours **ECC202** Max. Marks - 20

Note: - Missing data/ information if any, may be suitably assumed & mentioned in the answer. All questions carry equal marks [4=2+2].

	Question	CO
1a	Determine I_B , I_C , I_E , V_{CE} and V_{BE} in the circuit and region of operation. Assume that: $\beta = 150$ and $V_{BE} = 0.7V$.	CO ₁
1b	Draw and explain input and output V-I characteristics of common emitter (CE) configuration of BJT. Also derive the relationship between β and α .	CO ₁
2a	A 200 V, 50 Hz AC supply is connected to a half-wave rectifier with a load resistance of 2 k Ω and transformer turn ratio 10:1. Calculate DC Load Current (I_{dc}), rms current (I_{rms}), ripple factor and efficiency of the half-wave rectifier. Assume ideal diode.	CO ₁
2b	With suitable derivation, compare the efficiency of a bridge rectifier with a half-wave rectifier and justify why the bridge rectifier is preferred.	CO ₂
3.a	In electronics circuit a germanium PN junction diode is used which carries a current of 10.25 mA when it is forward biased with 0.25 V at $T = 300K$.	CO ₂



3b	For the given Zener diode regulator circuit, determine the load voltage V_L , load current I_L , Zener current I_Z , and series resistor current I_R when the load resistance $R_L = 1k\Omega$. Also, determine the minimum value of R_L required to ensure that the Zener diode operates in the breakdown (ON) region.	CO ₄
4	Identify the given circuits, draw the input waveform, output waveform, and the voltage transfer characteristic (VTC). Clearly indicate the voltage levels and explain the operation of the circuit during positive and negative half cycles.	CO ₃
5a		CO ₃
5b	Show Fermi levels in extrinsic and intrinsic semiconductor. Also, calculate the hole concentration in a silicon crystal with a donor concentration of $1.4 \times 10^{21} m^{-3}$, given that the intrinsic carrier concentration is $1.4 \times 10^{16} m^{-3}$. Further, determine the ratio of electron concentration to hole concentration.	CO ₄

